Scientific Computing & Advanced Programming Lab

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Lecture-Slides-6

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Gaussian Elimination

- We want to solve

$$A\mathbf{x} = \mathbf{b},$$

where

$$A \in \mathbb{R}^{n\times n}, \qquad \mathbf{x}, \mathbf{b} \in \mathbb{R}^n.$$

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Goal: Transform $[A \mid \mathbf{b}] \quad \longrightarrow \quad [U \mid \mathbf{c}]$, where U is upper triangular:

$$U = \begin{pmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ 0 & u_{22} & \cdots & u_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & u_{nn} \end{pmatrix}$$. Then solve $U\mathbf{x} = \mathbf{c}$

using **back substitution**.

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- This makes $a_{ik} = 0$.
- Entries below the pivot are eliminated.

Gaussian Elimination: Pseudocode

```
GaussianElimination(A, b)
    n = number of rows in A
    for k = 0 to n-2
        for i = k+1 to n-1
            m = A[i,k] / A[k,k]
            for j = k to n-1
                A[i,j] = A[i,j] - m*A[k,j]
            b[i] = b[i] - m*b[k]
    return A, b
```

The algorithm transforms

$$[A \mid \mathbf{b}] \quad \longrightarrow \quad [U \mid \mathbf{c}].$$

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Back Substitution

After forward elimination we have

$$U\mathbf{x} = \mathbf{c}.$$

For example,
$$\begin{aligned} u_{11}x_1 + u_{12}x_2 + u_{13}x_3 &= c_1, \\ u_{22}x_2 + u_{23}x_3 &= c_2, \\ u_{33}x_3 &= c_3. \end{aligned}$$

Start with the last equation and work upwards.

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For $i = n-1, n-2, \ldots, 0,$
$$\boxed{x_i = \frac{c_i - \displaystyle\sum_{j=i+1}^{n-1} u_{ij}x_j}{u_{ii}}}$$

$$\boxed{x_{n-1} \to x_{n-2} \to \cdots \to x_1 \to x_0}$$

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Back Substitution: Pseudocode

```
BackSubstitution(U, c)
    n = number of rows
    create x = zeros(n)
    for i = n-1 down to 0
        s = 0
        for j = i+1 to n-1
            s = s + U[i,j] * x[j]
        x[i] = (c[i] - s) / U[i,i]
    return x
```

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- try: Example:

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- The first pivot is zero. Swap the rows:

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 1 & 1 \end{array}\right].$$

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- Then swap:

$$R_k \leftrightarrow R_p.$$

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Choose the largest available pivot in the column.

Gaussian Elimination with Partial Pivoting

```
Input: A, b
n = number of rows
FOR k = 0, ..., n-2:

    # Find pivot
    p = index of largest |A[i,k]|,
        i = k, ..., n-1

    # Swap rows
    swap rows k and p of A
    swap b[k] and b[p]

    # Eliminate below pivot
    # continue as usual
```

From Gaussian Elimination to LU

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Key idea: Instead of throwing these multipliers away, store them in a lower triangular matrix L.

$$\boxed{A = LU}$$

Where Does L Come From?

Consider one elimination step:

$$R_i \leftarrow R_i - m_{ik} R_k.$$

The multiplier

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$$L = \begin{pmatrix} 1 & 0 & 0 \\ m_{21} & 1 & 0 \\ m_{31} & m_{32} & 1 \end{pmatrix}, \qquad U = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix}.$$

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Thus, Gaussian elimination gives us both:

$\boxed{U \text{ from the row operations}}$ $\boxed{L \text{ from the multipliers}}$

Why Factorize A?

Once $A = LU$, the system $A\mathbf{x} = \mathbf{b}$ becomes

$$LU\mathbf{x} = \mathbf{b}.$$

Introduce

$$L\mathbf{y} = \mathbf{b}.$$

Then solve

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So one system becomes two triangular systems:

$$\boxed{A\mathbf{x} = \mathbf{b} \quad \Longrightarrow \quad \begin{cases} L\mathbf{y} = \mathbf{b}, \\ U\mathbf{x} = \mathbf{y}. \end{cases}}$$

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Advantage: If A is fixed and many different $\mathbf{b}$'s are given, we factorize A only once.

LU Decomposition: Pseudocode

Input: Matrix $AU \leftarrow A, \qquad L \leftarrow I$

```
FOR k = 0,...,n-2:

    FOR i = k+1,...,n-1:

        m = U[i,k] / U[k,k]

        L[i,k] = m

        U[i,k:] = U[i,k:] - m U[k,k:]

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$$\boxed{A = LU}$$

Using LU to Solve $A\mathbf{x} = \mathbf{b}$

Suppose

$$A = LU.$$

Then

$$A\mathbf{x} = \mathbf{b} \quad \Longrightarrow \quad LU\mathbf{x} = \mathbf{b}.$$

Introduce an intermediate vector $\mathbf{y}$:

$$L\mathbf{y} = \mathbf{b}.$$

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$$\boxed{\begin{array}{ll} L\mathbf{y} = \mathbf{b} & \text{(forward substitution)} \\ U\mathbf{x} = \mathbf{y} & \text{(back substitution)} \end{array}}$$

LU: Solving a System

Consider

$$A = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 5 \\ 11 \end{pmatrix}.$$

LU decomposition gives

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Step 1: Forward substitution

$$L\mathbf{y} = \mathbf{b} \quad \Rightarrow \quad \begin{cases} y_1 = 5, \\ 2y_1 + y_2 = 11 \end{cases} \quad \Rightarrow \mathbf{y} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}.$$

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Step 2: Back substitution

$$U\mathbf{x} = \mathbf{y} \quad \Rightarrow \quad \begin{cases} 2x_1 + x_2 = 5, \\ x_2 = 1 \end{cases}$$